

HINTS AND SOLUTIONS

Q.1 B

Sol. $V = kT^{2/3}$

$$PV = 1. RT$$

$$W = \int p dV$$

$$dV = K \frac{2}{3} T^{-1/3} dT \quad \dots (1)$$

$$pkT^{2/3} = 1. RT$$

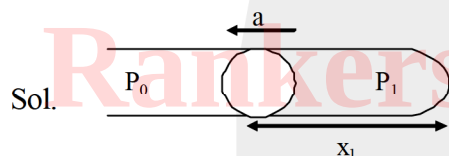
$$pk = RT^{1/3} \quad \dots (2)$$

from (1) & (2),

$$W = \int \frac{R}{k} T^{1/3} \cdot k \frac{2}{3} T^{-1/3} dT$$

$$W = \frac{2R}{3} \int 1 dT = \frac{2R}{3} \times 30 = 20R$$

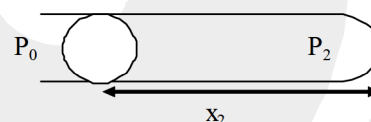
Q.2 D



$$P_1 = P_0 + \frac{a}{g} l = 72.5 + 2.5 = 75 \text{ cm of Hg}$$

$$P_2 = 72.5 - 2.5 = 70 \text{ cm of Hg}$$

$$P_2 x_2 = P_1 x_1 \Rightarrow x_2 = \frac{75 \times 20}{70} = 21.42 \text{ cm}$$



Q.3 C

Sol. $P_h = P_0 e^{-\frac{mgh}{kT}}$

$$P_0 = \frac{m_\infty g}{A}$$

$\begin{cases} m_\infty = \text{mass till } \infty \text{ of column of area } A \\ m_h = \text{mass till height } h \text{ of column of area } A \end{cases}$

$$P_h = \frac{(m_\infty - m_h)g}{A}$$

$$h \frac{(m_\infty - m_h)g}{A} = \frac{m_\infty g}{A} e^{-\frac{mgh}{kT}}$$

$$1 - \frac{m_h}{m_\infty} = e^{-\frac{mgh}{kT}}$$

$$\frac{m_h}{m_\infty} = \frac{n_h}{n_\infty} = 1 - e^{-\frac{mgh}{kT}}$$

Q.4 B

Sol. $\Delta U = mS_v \Delta T$

Q.5 C

Q.6 B

Sol. $W = \int_{V_i}^{V_f} P dV$ lin equation $P = \left[\frac{3P_0}{4V_0} V + \frac{9P_0}{4} \right]$

$$W = \int_{V_0}^{5V_0} \left(\frac{3P_0}{4V_0} V + \frac{9P_0}{4} \right) dV$$

$$W = \frac{3P_0}{4V_0} \left(\frac{V^2}{2} \right)_{V_0}^{5V_0} + \frac{9P_0}{4} [V]_{V_0}^{5V_0}$$

$$W = 18 P_0 V_0$$

$$\Delta U = \frac{P_f V_f - P_i V_i}{(5/3 - 1)} = \frac{30P_0 V_0 - 3P_0 V_0}{2/3} = \frac{81 P_0 V_0}{2}$$

$$\Delta Q = W + \Delta U = \frac{117}{2} P_0 V_0$$

$$nC (T_f - T_i) = \frac{117}{2} P_0 V_0$$

$$C \left(\frac{6P_0 \times 5V_0}{R} - \frac{3P_0 V_0}{R} \right) = \frac{117}{2} P_0 V_0$$

$$C = \frac{\frac{117}{2} P_0 V_0}{\frac{27P_0 V_0}{R}} = \frac{13R}{6}$$

$$C = \frac{13R}{6}$$

Q.7 B

Q.8 B

Sol. $w = P_1 \Delta V - P_0 \Delta V$
 $= (P_1 - P_0) (V_1 - V_0)$
 $= P_1 V_1 + P_0 V_0 - P_0 V_1 - V_0 P_1$
 $= nRT_3 + nRT_1 - 2nRT_2$
 $P_1 V_1 = nRT_3$
 $P_0 V_0 = nRT_1$
 $P_1 V_0 = nRT_2$
 $P_0 V_1 = nRT_2$
Hence $P_1 V_1 P_0 V_0 = nRT_3 nRT_1$
 $P_1 V_0 P_0 V_1 = nRT_2 nRT_2$
 $T_1 T_3 = T_2^2$

$$= R[T_3 + T_1 - 2\sqrt{T_1 T_3}]$$

Q.9 C

Sol. Process AB and CD are isothermal

$$\therefore \Delta W = nRT \ln\left(\frac{V_f}{V_i}\right)$$

Dotted lines are isochoric lines as they pass through origin

$$\therefore \frac{V_f}{V_i} \text{ is same for both process}$$

$$\therefore \Delta W_{A \rightarrow B} = nRT_1 \ln\left(\frac{V_f}{V_i}\right)$$

$$\Delta W_{C \rightarrow D} = nRT_2 \ln\left(\frac{V_f}{V_i}\right)$$

$$\text{So, } T_1/T_2 = 2$$

Q.10 A

Sol. For isothermal process

$$Q = W = nRT \ln \frac{V_f}{V_i}$$

$$15000 = (0.5) \left(\frac{25}{3}\right) T \ln \frac{0.3}{0.1}$$

$$T = 360 \text{ K}$$

Q.11 B

Q.12 D

Q.13 D

Q.14 C

$$\text{Sol. Area of triangle} = \frac{1}{2} (\text{Base}) \times \text{Height}$$

$$= \frac{1}{2} 8P_A 2V_A = 8P_A V_A$$

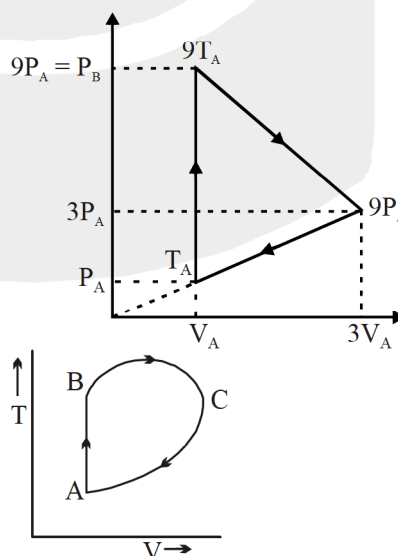
$$= 8P_A V_A$$

$$\begin{aligned} \text{For BC P-V eq. is } P &= aV + b \\ 9P_A &= aV_A + b \\ 3P_A &= 3aV_A + b \end{aligned}$$

$$\Rightarrow a = -\frac{6P_A}{2V_A} = -3\frac{P_A}{V_A}$$

$$\begin{aligned} b &= 9P_A - aV_A \\ &= 9P_A + 3P_A \\ b &= 12P_A \end{aligned}$$

$$\therefore P = -\frac{3P_A}{V_A} V + 12P_A$$



$$PV = -\frac{3P_A}{V_A} V^2 + 12P_A V$$

$$nRT = -3P_A/V_A V^2 + 12P_A V$$

$$nR \frac{dT}{dV} = -\frac{3P_A}{V_A} (2V) + 12P_A$$

$$dT/dV = 0 \quad \frac{6P_A}{V_A} V = 12P_A \Rightarrow V = 2V_A$$

By equation (1)

$$nRT = -\frac{3P_A}{V_A} (2V_A)^2 + 12P_A (2V_A)$$

$$= -12P_A V_A + 24P_A V_A$$

$$nRT = 12P_A V_A$$

$$nRT = 12nRT_A$$

$$T = 12T_A$$

$$= 12 \times 200$$

$$= 2400 \text{ K}$$

Q.15 B

Q.16 ABCD

Q.17 BCD

Q.18 AC

Sol: $Q_{123} = 100$ $W_{123} = 40$

$$W_{143} = 10$$

$$\text{Hence } V_{123} = 100 - 40 = 60$$

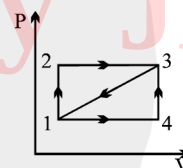
$$Q_{143} = 60 + 10 = 70$$

$$W_{31} = 10 + \frac{40 - 10}{2} = -25 \text{ cal}$$

$$Q_{31} = -25 - 60 = -85$$

(A)

(C)



Q.19 BD

Sol. For the process (i) $PV^2 = \text{constant}$

$$\left(\frac{nRT}{V}\right) V^2 = \text{constant}$$

$$\therefore T \propto \frac{1}{V}$$

for the process (ii) $P = KV^2$

$$\therefore T \propto V^3$$

$\therefore$ On increasing volume, final temperature in process (ii) is greater than that in process (i)

$$\text{Now, for process } PV^m = \text{constant, molar heat capacity, } C = \frac{R}{\gamma - 1} - \frac{R}{m - 1}$$

$$\therefore \text{ for process (i) } C_1 = C_v - R$$

$$\text{for process (ii) } C_2 = C_v - \frac{R}{(-2 - 1)} = C_v + \frac{R}{3}$$

$$C_2 > C_1 \text{ \& } (\Delta T)_2 > (\Delta T)_1$$

$$\therefore Q_2 > Q_1]$$



Q.20 [Ans. (A)-P, (B)-S, (C)-R, (D)-Q]

Sol. $w_{iaf} + u_{if} = H_{iaf}$
 $20 + u_{if} = 50$
 $u_{if} = 30$

and $w_{ibf} + u_{if} = H_{ibf}$
 $w_{ibf} + 30 = 36$
 $w_{ibf} = 6$

(A) $w_{ibf} = 6$

(B) $H_{fi} = w_{fi} + u_{fi}$
 $= 13 + 30 = 43$

(C) $u_{if} = 30$ $u_i = 10$ $u_f = 40$

(D) $H_{ib} = w_{ib} + u_{ib}$
 $= 6 + (22 - 10) = 18$

Q.21 [Ans. (A) P,Q,R (B) P,R,S (C) P,Q (D) T]

Sol. (A) $B \rightarrow C$ is isobaric expansion
 $T_B < T_C$ and $V_B < V_C$
 So $\Delta U > 0$ and $\Delta W > 0$; Hence $\Delta Q > 0$

(B) $A \rightarrow B$ is isochoric heating
 $T_B > T_C$ and $V_B = V_A$
 So $\Delta U > 0$ and $\Delta W = 0$; Hence $\Delta Q > 0$

(C) $A \rightarrow B \rightarrow C \rightarrow A$ is cyclic process
 $\Delta T = 0$ So, $\Delta U = 0$ But $\Delta W > 0$ and Hence $\Delta Q > 0$

(D) $C \rightarrow A$
 $T_A < T_C$ and $V_A < V_C$
 So $\Delta U < 0$ and $\Delta W < 0$, So $\Delta Q < 0$

Q.22 [Ans. 0008]

Sol. $P = \frac{nRT}{V} \Big|_{P_{\max}} = \frac{nR(4T)}{V} \Big|_{P_{\min}} = \frac{nRT}{2V}$
 $P_{\max} / P_{\min} = 8$

Q.23 [Ans. 0005]

Sol. $_{vss}$ Temperature is constant $\Rightarrow \Delta E = 0, dW = nRT \frac{dv}{v} = nRT \frac{Adx}{AL/2}$

$Q = \Delta E + W$

$\Rightarrow \frac{dW}{dt} = \frac{nRT}{L/2} \frac{dx}{dt}$

$\frac{dQ}{dt} = \frac{dW}{dt} \Rightarrow k \frac{1}{900} \frac{\Delta T}{L} = \frac{2nRT}{L} \left(\frac{dx}{dt} \right)$

$\Rightarrow \frac{dx}{dt} = \frac{k \times 27}{900 \cdot nRT} = \frac{415.5 \times 27 \times 2}{900 \times 0.5 \times 8.31 \times 300} = \frac{1}{200} \text{ m/s} = 5 \text{ mm/s}$